

Outbreak: Riverton

AP Biology · public health

A hospital campus in week three: triage marquees, container labs, and a fence with one gate.

45	7	6.6	3/7	2	12 · 32 · 1
STOPS	CHOICE (16%)	EFFECTIVE FORMATS	EQUATIONS COMPUTED	SELECT-ONLY CONCEPTS	SELECT · CONSTRUCT · OPERATE

READING Where this campaign stands

Two of the four gaps are not gaps but absences: $R_e = R_0 \times S$ and $CFR = \text{deaths} / \text{cases}$ are mentioned at no stop at all. An AP Biology public-health course that never reaches the effective reproduction number is missing its spine. The mix is healthy — SEQUENCE-led, twelve of them — so this campaign’s work is almost entirely curriculum.

NOW The mix as it ships

SEQUENCE 12	CHOICE 7	BALLPARK 7	DIAGNOSIS 5	PROTOCOL 5	VALUE 2	ALLOCATE 2	CONTROL 1	TRACE 1	CHAIN 1	TRIGGER 1	ATTEST 1
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CURRICULUM What the syllabus is not getting

EQUATION THE COURSE OWES	WHAT IT CARRIES	MENTIONED AT	STATUS
$R_e = R_0 \times S$	effective reproduction number against a susceptible fraction	– no stop at all	real gap
$PPV = (\text{prev} \cdot \text{sens}) / (\text{prev} \cdot \text{sens} + (1 - \text{prev})(1 - \text{spec}))$	why a good test fails at low prevalence	15 · BALLPARK 22 · PROTOCOL 26 · SEQUENCE	real gap
$CFR = \text{deaths} / \text{cases}$	case fatality, and what its denominator hides	– no stop at all	real gap
$pH = pK_a + \log([A^-] / [HA])$	Henderson–Hasselbalch — buffering in a body	6 · CHOICE 18 · BALLPARK	real gap

0 of 4 sit on an instrument stop, whose board arithmetic the coverage check cannot yet read — expect Phase 0 to clear those. 4 are unaffected by it.

TIER Concepts the player only ever picks from a list

#	MECHANISM CONCEPT	STOPS
3	Enzymes and metabolism	6
23	Gene regulation and expression control	24

Reported, never failed. Select-tier can be exactly right; these are the candidates, not the defects.

FUN What this subject and this place can carry

TALLY STRONG

a POP stop

Cases accumulate into bins and the statistic moves because there is not enough behind it yet, not because the epidemic changed. That confusion is the single most consequential error in outbreak reporting, and it is what TALLY grades.

SERVES R_e and CFR — the two equations the game does not have

NEEDS a run where an early batch flatters the number, and a pass mark that stays unprinted

TRIAL STRONG

the decon tunnel

The place already has the lesson built into it: courtyards, triage marquees, container labs, a decon tunnel on the main route and a fence with one gate. Driving that route in the wrong order is how a campus infects itself.

SERVES transmission and barrier practice; the place stops being scenery

RISK gates stood off each door by $d/2 + 10$, or every ring renders under the floor

SPOT STRONG

CLIN or the marquees

A triage criterion replaced mid-shift with nothing announcing it. This is not a game conceit — it is week three of an outbreak, and the case definition changes.

SERVES case definition, and concept 11 — homeostasis and feedback, currently select-only

RISK score only the discriminating items

BELT STRONG

the one gate

A fence with one gate is a sorting line whether anybody designed it as one. Sort arrivals against the case definition at the speed people actually arrive.

SERVES case definition as one bit, held at speed

RISK must be decidable at a glance; a definition needing two clauses is not a BELT

PROBE MEDIUM

FIELD · One Health

Sample ward by ward or village by village and find where the chain breaks. Strong on subject; medium only because the campus is compact and the chain would be diagrammatic rather than walked.

SERVES reservoir and transmission chain

HOLD DEFER

Isolation-room pressure held against a closing band is real hospital engineering and thin biology. Defer.

LOB REJECT

Nothing here is aimed.

SLATES Three appetites — the campaign is zero-sum

No conversion adds a stop. Each changes what an existing stop **does** while leaving what it **teaches** alone. Snapshot before, diff after: `curriculumDelivery.mjs outbreak_riverton --snapshot before.json`, then `--against before.json`.

Minimum honest 3 STOPS · ONE SITTING

15	d5	POP	BALLPARK → TALLY	What does a positive mean? accumulate tests until a positive means something — and PPV with it
24	d8	CELL	CHOICE → CHAIN	The same genome, different cells the same genome, different cells: name the step that governs expression
6	d2	MOL	CHOICE → HOLD	When heat changes an enzyme assay the enzyme assay held in its band while the heat rises

CHOICE 7 → 5 (11%) · moves 12·32·1 → 10·32·3 · effective formats 6.6 → 7.4 · 2 new operate stops, 1 of them arcade

Balanced 6 STOPS · THE RECOMMENDATION

15	d5	POP	BALLPARK → TALLY	What does a positive mean?
24	d8	CELL	CHOICE → CHAIN	The same genome, different cells
6	d2	MOL	CHOICE → HOLD	When heat changes an enzyme assay
22	d8	IMM	PROTOCOL → SPOT	Match the surveillance signal a surveillance signal under a case definition that changes
26	d9	FIELD	SEQUENCE → PROBE	Test the reservoir hypothesis the reservoir found station by station rather than hypothesised
18	d6	CLIN	BALLPARK → BALLPARK	Read the downstream acid–base signal the acid–base signal computed, not described

CHOICE 7 → 5 (11%) · moves 12·32·1 → 10·30·5 · effective formats 6.6 → 8.3 · 4 new operate stops, 2 of them arcade

Ambitious 9 STOPS · ADDS WORLD WORK

15	d5	POP	BALLPARK → TALLY	What does a positive mean?
24	d8	CELL	CHOICE → CHAIN	The same genome, different cells
6	d2	MOL	CHOICE → HOLD	When heat changes an enzyme assay
22	d8	IMM	PROTOCOL → SPOT	Match the surveillance signal
26	d9	FIELD	SEQUENCE → PROBE	Test the reservoir hypothesis
18	d6	CLIN	BALLPARK → BALLPARK	Read the downstream acid–base signal
4	d2	CLIN	DIAGNOSIS → TRIAL	What kind of agent fits the whole panel? the decon route, driven
9	d3	CELL	CONTROL → BELT	Separate binding from entry the gate, sorting against the definition in force
30	d10	POP	BALLPARK → CLOUD	Track a rising variant a case–fatality estimate as a band, not a number

CHOICE 7 → 5 (11%) · moves 12·32·1 → 9·30·6 · effective formats 6.6 → 9.0 · 6 new operate stops, 4 of them arcade

THE ORDER OF WORK

1. Phase 0 — teach the coverage check to read an instrument's own board, and re-baseline curriculum–debt.json. Rewriting a stop that already teaches the thing is the exact failure the check exists to prevent. 2. Snapshot. 3. Edit the book, never the generated content — tools/import-book.mjs books/outbreak-riverton.yml outbreak_riverton --verify. 4. npm run traps: every instrument's trap must fire. 5. npm run check outbreak_riverton and npm run drive outbreak_riverton, right answer and wrong — ask what a player who understood nothing would score before believing any pass mark. 6. --against the snapshot. A changed format is reported; a changed objective fails.

Counts computed from `themes/outbreak_riverton/content/` via `engine/dev/curriculumDelivery.mjs`; concept and equation lists authored in `tools/syllabus.js`. Format verdicts and slates are editorial. Run it with `THEME=outbreak_riverton npm run dev`.